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(54) **POROUS MEMBRANE, BATTERY SEPARATOR OBTAINED USING SAME, AND METHOD OF PRODUCING SAME**

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See application file for complete search history.

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(57) **ABSTRACT**

Considering that the battery separator of embodiments of the present invention will have thinner materials and lower costs, provided are a polyolefin porous membrane with exceptionally high peel strength between the polyolefin porous membrane and a modified porous layer, suitable for high-speed processing during slit processing and the battery assembly process, and suitable for laminating on a modified porous layer, and a battery separator obtained by laminating a modified porous layer on the polyolefin porous membrane. A polyolefin porous membrane comprising protrusions of polyolefin having a size (W) within a range of 5 μm≤W≤50 μm and a height (H) within a range of 0.5 μm≤H and irregularly disposed on both sides of the polyolefin porous membrane in a density not less than 3/cm^2 and not more than 200/cm^2 per side, and the polyolefin porous membrane having a thickness of not more than 25 μm.

8 Claims, 2 Drawing Sheets

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B32B 27/20	(2006.01)
B32B 3/30	(2006.01)
B32B 15/04	(2006.01)
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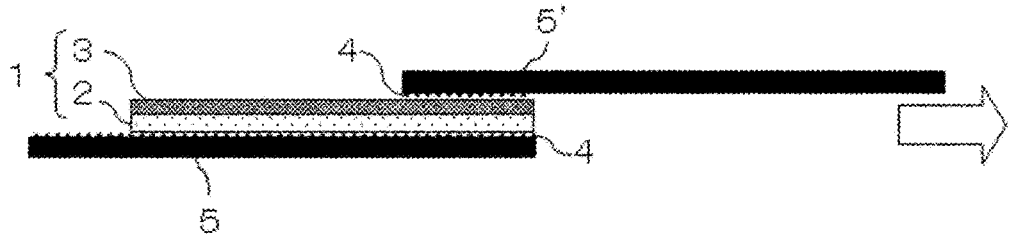


FIG. 1

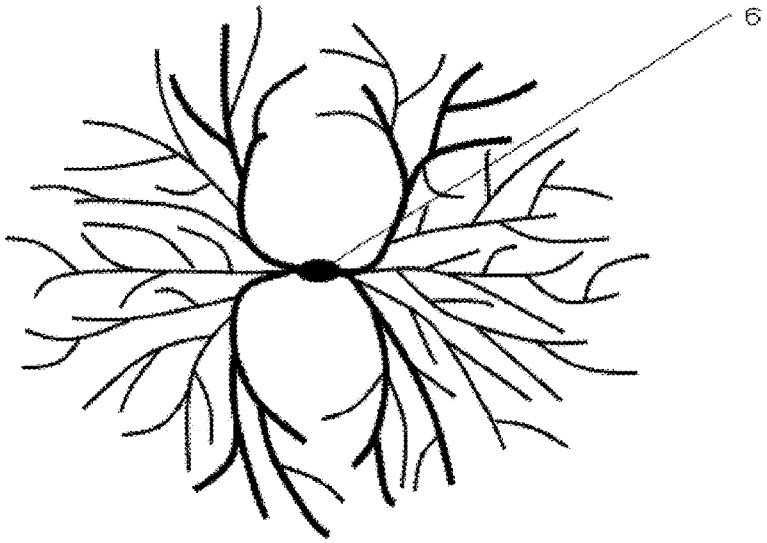


FIG. 2

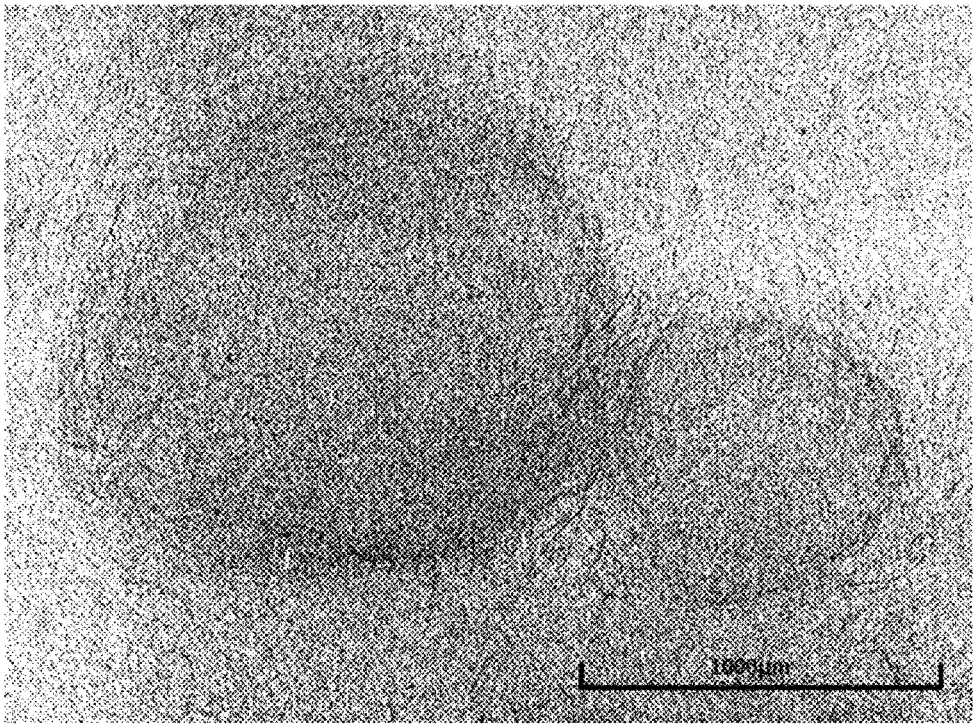


FIG. 3

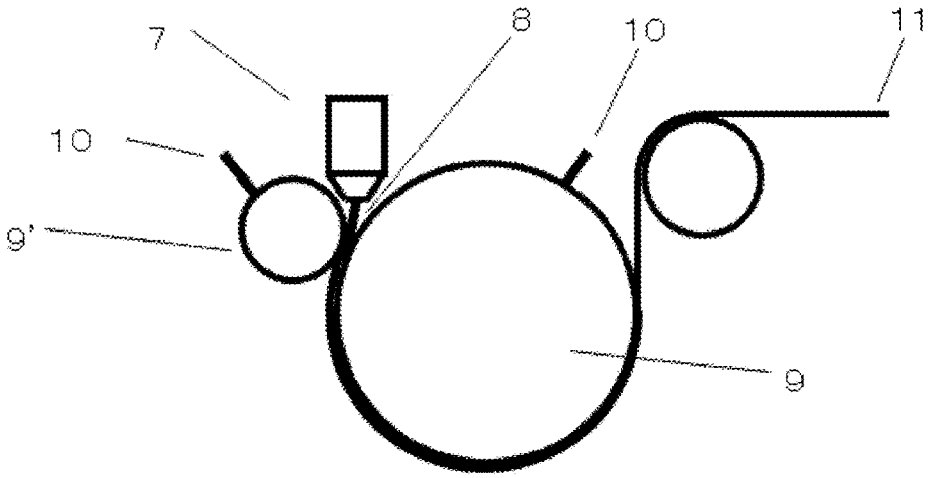


FIG. 4

POROUS MEMBRANE, BATTERY SEPARATOR OBTAINED USING SAME, AND METHOD OF PRODUCING SAME

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a United States National Phase Application of co-pending International Patent Application Number PCT/JP2014/075828, filed Sep. 29, 2014 and entitled "POLYOLEFIN POROUS FILM, SEPARATOR FOR BATTERIES WHICH IS MANUFACTURED USING SAID POROUS FILM, AND METHODS RESPECTIVELY FOR MANUFACTURING SAID POROUS FILM AND SAID SEPARATOR", which Application claims priority to Japanese Patent Application Number 2013-208461, filed with the Japanese Patent Office on Mar. 10, 2013, the contents of which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

The present invention relates to a polyolefin porous membrane that is suitable for laminating a modified porous layer, and to a battery separator comprising the polyolefin porous membrane.

BACKGROUND

Thermoplastic resin microporous membranes are used widely as filters and separators.

Specifically, they are used in battery separators for lithium ion rechargeable batteries, nickel-metal hydride batteries, nickel-cadmium batteries, or polymer batteries, separators for electric double layer capacitors, various filters such as ultrafiltration membranes, microfiltration membranes, and the like, moisture permeation waterproof clothes, medical materials and the like.

In particular, when used as a separator for a lithium ion rechargeable battery, with the increasing capacity and power and decreasing weight of batteries, the thermoplastic resin microporous membrane requires mechanical characteristics as well as physical properties such as heat resistance, permeability, dimensional stability, pore blocking characteristics, membrane melt-puncture characteristics, electrical insulating properties, electrolytic solution resistance, oxidation resistance, and the like.

Additionally, polyolefin porous membranes have been advantageously used as separators for lithium ion rechargeable batteries due to having ion permeability due to electrolytic solution impregnation and possessing not only excellent electrical insulating properties, electrolytic solution resistance, and anti-oxidation properties, but also a pore blocking effect, which blocks the electrical current to prevent excessive temperature increases in the temperature range of about 120 to 150° C. when battery temperature increases abnormally.

On the other hand, if the temperature continues to increase even after pore blocking, the membrane may puncture due to a decrease in viscosity of the polyethylene that constitutes the membrane and shrinkage of the membrane.

Furthermore, polyolefin porous membranes further require improved adhesion to electrode materials for improvement in cycle characteristics of batteries and improved wettability toward electrolytic solution for productivity improvement.

To solve these problems, lamination of various modified porous layers on a porous membrane has been studied.

As modified porous layers, polyamideimide resin, polyimide resin, and polyamide resin, which have both good heat resistance and good wettability toward electrolytic solution, fluorine-based resin, which exhibits good adhesion toward electrode, and the like are preferably used.

A modified porous layer refers to a layer that includes a resin that provides or improves at least one of the functions among heat resistance, adhesion to an electrode material, wettability toward electrolytic solution, and the like.

Furthermore, it is necessary to increase the area not only of the electrode but also of the separator to fill a container in order to increase battery capacity, and a decrease in thickness of the separator is expected.

However, since the thinner porous membrane may experience deformation in its planar directions, a modified porous layer laminated on a thin porous membrane in a battery separator may delaminate during processing, slit processing, or the battery assembly process, and the safety may be compromised.

Additionally, it is anticipated that the battery assembly process will be sped up for the purpose of reducing cost.

To obtain a separator that has few problems, such as delamination of the modified porous layer even in high-speed processing, a polyolefin porous membrane having high adhesion with a modified porous layer that can withstand high-speed processing is required.

However, when the resin included in the modified porous layer is sufficiently permeated into the polyolefin porous membrane in order to improve the adhesion, large increases in air permeation resistance have been problematic.

Patent Document 1 discloses that a solution containing polyvinylidene fluoride is coated on one side of a 9 μm thick polyethylene porous membrane, and due to a fraction of polyvinylidene fluoride resin adequately penetrating into fine pores of the polyethylene porous membrane, it exhibits an anchor effect. As a result, a composite porous membrane with a peel strength (T-peel strength) of 1.0 to 5.3 N/25 mm at the interface between the polyethylene porous membrane and the polyvinylidene fluoride coating layer is obtained.

Patent Document 2 discloses that a heat-resistant porous layer, which contains acrylic resin, an N-vinyl acetamide polymer or a thickener of water soluble cellulose derivative, and plate-like boehmite, is disposed on a 16 μm thick, corona-discharge treated polyethylene porous membrane, resulting in a separator with a 180° peel strength (T-peel strength) of 1.1 to 3.0 N/10 mm at the interface between the polyethylene porous membrane and the heat-resistant porous layer.

Patent Document 3 discloses a method of producing a porous membrane, in which a polyethylene solution that comprises 30 parts by weight of a polyethylene composition (comprising 20 wt. % of ultrahigh molecular weight polyethylene (UHMWPE) with a weight average molecular weight of 2.0×106, 80 wt. % of high density polyethylene (HDPE) with a weight average molecular weight of 3.5×105, and antioxidant) and 70 parts by weight of liquid paraffin are extruded from an extruder at 190° C., and the extrudate is drawn by a chill-roll kept at 50° C. The resultant gel-like product is stretched biaxially to 500%×500%, and a porous membrane is obtained.

In Patent Document 4, the method of producing a microporous membrane is disclosed, in which polyethylene solution similar to that in Patent Document 3 is extruded from an extruder, and the extrudate is drawn by a chill-roll kept at 0° C. The resultant gel-like product is stretched biaxially to 500%×500%, and the microporous membrane is obtained.

In Patent Document 5, Working Example 1 discloses that a polyethylene solution that comprises 50 parts by mass of a composition (comprising 47.5 parts by mass of polyethylene with a viscosity-average molecular weight of 200,000, 2.5 parts by mass of polypropylene with a viscosity-average molecular weight of 400,000, and antioxidant) and 50 parts by mass of liquid paraffin is extruded from an extruder at 200° C., and the extrudate is drawn by a chill-roll kept at 25° C. to obtain a gel-like product, which is then biaxially stretched to 700%×640%, and a polyolefin resin porous membrane is obtained. A coating layer comprising polyvinyl alcohol and alumina particles is laminated on one side of the obtained polyolefin resin porous membrane, to yield a multi-layer porous membrane.

In Patent Document 6, Working Example 6 discloses that a polyethylene solution that comprises 30 wt. % of a polyethylene composition (comprising polyethylene of weight average molecular weight 4,150,000 and polyethylene of weight average molecular weight 560,000 in a weight ratio of 1:9) and 70 wt. % of solvent mix of liquid paraffin and decalin is extruded from an extruder at 148° C. and the extrudate is cooled in a water bath. The resultant gel-like product is then stretched biaxially to 550%×1,100%, and a polyethylene porous membrane is obtained. A coating layer comprising meta-type wholly aromatic polyamide and alumina particles is laminated on both sides of the obtained polyethylene resin porous membrane, to yield a non-water-based separator for a rechargeable battery.

Patent Document 7 discloses a polyolefin microporous membrane with an embossed pattern of diagonal lattices obtained by passing a gel sheet between an embossing roll and a back-up roll before stretching.

However, the separators described in Patent Documents 1 to 7 will not sufficiently ensure safety due to partial delamination of the modified porous layers during slit processing or battery assembly process, when it is required that the manufacturing processes are sped up and thinner separators are provided accompanying the lower cost and higher capacity in the near future.

In particular, as the polyolefin resin porous membrane that serves as a substrate becomes thinner, it becomes more difficult to ensure safety sufficiently since it is more difficult to achieve enough anchoring of modified porous layers to the polyolefin resin porous membrane.

CITATION LIST

Patent Literature

Patent Document 1: Japanese Unexamined Patent Application Publication No. 2012-043762A

Patent Document 2: WO/2010/104127

Patent Document 3: Japanese Unexamined Patent Application Publication No. 2003-003006A

Patent Document 4: Japanese Unexamined Patent Application Publication No. 2004-083866A

Patent Document 5: Japanese Unexamined Patent Application Publication No. 2009-26733A

Patent Document 6: WO/2008/149895

Patent Document 7: WO/2008/053898

SUMMARY OF INVENTION

Technical Problem

Considering that battery separators will require thinner materials and lower costs in the future, an object of embodiments of the present invention is to provide a polyolefin porous membrane with exceptionally high peel strength against a modified porous layer, suitable for high-speed processing during slit processing and the battery assembly process, and suitable for laminating on a modified porous layer, and a battery separator obtained by laminating a modified porous layer on the polyolefin porous membrane.

Peel strength between a polyolefin porous membrane and a modified porous layer referred in the present specification is a value obtained from the measurement described below (in some cases, it is also referred to as 0° peel strength hereinafter).

FIG. **1** is a schematic side view of a laminated sample of a polyolefin porous membrane and a modified porous layer under tension provided by a tensile tester (not illustrated).

The reference numeral 1 denotes a laminated sample, 2 denotes a polyolefin porous membrane, 3 denotes a modified porous layer, 4 denotes a double-sided adhesive tape, 5 and 5' denotes aluminum plates, and the arrow in the figure indicates the direction of tensile force.

On the aluminum plate (**5**), 50 mm×25 mm in size and 0.5 mm in thickness, the double-sided adhesive tape (**4**) of the same size is adhered. The surface of the polyolefin porous membrane (**2**) side of the sample (**1**) (the battery separator), cut out in the size of 50 mm in width and 100 mm in length, is adhered on the aluminum plate (**5**), so that the 40 mm portion of the sample, measured from one of the 25 mm long edges of the aluminum plate (**5**), is overlapped and adhered together, while the protruded portion is cut out.

Next, a double-sided adhesive tape is adhered on one side of the aluminum plate (**5'**), 100 mm in length, 15 mm in width and 0.5 mm in thickness. And the 20 mm portion of the sample, measured from one of the 25 mm long edges of the aluminum plate (**5**), is overlapped and adhered together.

Then, using the tensile tester, the aluminum plate **5** is fixed, and the aluminum plate (**5'**) is pulled in parallel to the aluminum plate (**5**) at a tensile velocity of 10 mm/min, and the strength when the modified porous layer is delaminated is measured.

When the peel strength measured by this evaluation method is not less than 100 N/15 mm, then there will be few incidents of delamination of the laminated modified porous layer during the conveyance and the process, even if the thickness of the polyolefin porous membrane is not more than 10 μm, for example.

T-peel strength or 180° peel strength, both of which are conventionally used for peel strength measurement, is the peel force required to peel off the coating layer in the direction perpendicular or diagonally backward from perpendicular, relative to the polyethylene porous membrane surface.

Compared to these conventional evaluation methods, this evaluation method may provide the better evaluation in relation to the actual scratch resistance during slit processing and the battery assembly process.

Solution to Problem

In order to solve the problems described above, a polyolefin porous membrane of the present invention comprises the configuration below:

A polyolefin porous membrane comprising protrusions of polyolefin having size (W) within a range of 5 μm≤W≤50 μm and height (H) within a range of 0.5 μm≤H and irregularly disposed on both sides of the polyolefin porous membrane in a density not less than 3/cm^2 and not more than

200/cm^2 per side, a thickness of the polyolefin porous membrane being not more than 25 μm.

In a battery separator of the present invention, the modified porous layer is preferably laminated on at least one side of the polyolefin porous membrane.

Embodiments of the modified porous layer in the battery separator of the present invention preferably comprises polyamideimide resin, polyimide resin or polyamide resin.

Embodiments of the modified porous layer in the battery separator of the present invention preferably comprises fluorine-based resin.

Embodiments of the modified porous layer in the battery separator of the present invention preferably comprises carboxymethylcellulose (CMC) or acrylic-based resin.

Embodiments of the modified porous layer in the battery separator of the present invention preferably comprises inorganic particles or cross-linked polymeric particles.

The thickness of embodiments of the polyolefin porous membrane in the battery separator of the present invention is preferably not more than 20 μm.

The thickness of embodiments of the polyolefin porous membrane in the battery separator of the present invention is preferably not more than 16 μm.

In order to solve the problems described above, a method of manufacturing embodiments of the polyolefin microporous membrane of the present invention has the configuration below:

A method of manufacturing a polyolefin porous membrane, the method comprising the steps of:

(a) preparing polyethylene resin solution by melting and kneading after adding a forming solvent to polyethylene resin;

(b) extruding the polyethylene resin solution through a T-die, chilling by a chill-roll having a surface from which the forming solvent has been removed, and disposed on both sides of the polyethylene resin solution that has been extruded in membrane form, and forming a gel-like product;

(c) stretching the gel-like product in a machine direction and in a transverse direction to obtain a stretched and molded material;

(d) extracting and removing the forming solvent from the stretched and molded material followed by drying, to obtain a porous molded material;

(e) thermal treating the porous molded material to obtain a polyolefin porous membrane.

In embodiments of the method of manufacturing a polyolefin porous membrane of the present invention, a means to remove the forming solvent in step (b) is preferably a doctor blade.

Advantageous Effects of Embodiments of Invention

Embodiments of the polyolefin porous membrane of the present invention has the protrusions described above on both sides.

According to embodiments of the present invention, when a modified porous layer is provided on both sides of the polyolefin porous membrane, the adhesion between the polyolefin porous membrane and the modified porous layer is outstanding on both sides, and a battery separator in which delamination does not occur even in high-speed conveyance is obtained.

Furthermore, not only is adhesion outstanding, but it exhibits the effect that there is little change in gas permeation resistance even when stored as a roll for a long period, for example, one month or more, even when the modified porous layer is provided on only one side of the polyolefin porous membrane.

This is because, for example, when a separator having a polyolefin porous membrane on the inside and having a modified porous layer on the outside relative to the roll shaft is made into a roll of separator, space can be formed between the polyolefin porous membrane of the separator of the outside and the modified porous layer of the separator on the inside by means of protrusions on the polyolefin porous membrane, and as a result, the roll hardness is relatively low and structural deformation of the porous membrane by its own weight is suppressed.

BRIEF DESCRIPTION OF DRAWINGS

FIG. **1** is a schematic view of the measurement method for 0° peel strength.

FIG. **2** is a schematic view illustrating a spherulite structure and a spherulite nucleus of polyethylene in the polyethylene porous membrane.

FIG. **3** is a micrograph of a ring-shaped mark on the polyethylene porous membrane, originating from the spherulite of polyethylene.

FIG. **4** is a schematic view of the steps to extrude polyethylene resin solution through a T-die disposed at the end of the extruder and to chill the extrudate by a chill-roll to form a gel-like product.

DESCRIPTION OF EMBODIMENTS

Embodiments of the present invention is a polyolefin porous membrane having protrusions of an appropriate shape and number on both sides, which can be obtained by using a specific polyolefin resin solution and by controlling to a high degree the cooling rate of the polyolefin resin solution extruded through a T-die from an extruder during the steps of manufacturing the polyolefin porous membrane.

Furthermore, embodiments of the present invention is a polyolefin porous membrane, which, upon laminating a modified porous layer on the polyolefin porous membrane, has excellent peel strength against the modified porous layer, and exhibits a small increase in air permeation resistance.

The protrusions described in embodiments of the present invention are fundamentally different from those obtained by adding inorganic particles, for example, to a polyolefin porous membrane.

The height of the protrusions obtained by adding inorganic particles to a polyolefin porous membrane is usually very low, and forming the protrusions with height not less than 0.5 μm in the same manner would require particles with diameters equal to or larger than the thickness of the polyolefin porous membrane.

However, addition of such particles would decrease the strength of the polyolefin porous membrane and is unrealistic.

The protrusions described in embodiments of the present invention are obtained by growing the parts of the polyolefin porous membrane to appropriate shapes and do not degrade the basic properties of the polyolefin porous membrane.

"Irregularly disposed" in embodiments of the present invention indicates that the arrangement of protrusions is clearly different from the patterned or periodic arrangement obtained by the process of an embossment roll before or after the stretching process during the manufacturing of the polyolefin porous membrane.

In principle, press working such as embossment forms protrusions by compressing the parts other than the protrusions, and is not preferable due to possible decrease in air permeation resistance and in wettability toward electrolytic solution.

The protrusions with appropriate shape described in embodiments of the present invention are not less than 5 μm and not more than 50 μm in size, and are not less than 0.5 μm in height.

Namely, 5 μm≤W≤50 μm, where W is protrusion size and 0.5 μm≤H, where H is protrusion height.

Such protrusions can function as anchors upon laminating the modified porous layer on the porous membrane, and a battery separator with a large 0° peel strength described above can be obtained as a result.

Meanwhile, the upper limit of the height is not particularly limited, but the height of 3.0 μm may be sufficient.

Larger population of protrusions with sufficient height is likely to increase the 0° peel strength described above.

Namely, 0° peel strength is influenced by the number and the average height of protrusions with height not less than 0.5 μm.

The number of protrusions must be not less than 3/cm^2 on each of the sides.

The lower limit of the number of the protrusions is preferably 5/cm^2, and more preferably 10/cm^2.

The upper limit of the number of the protrusions is preferably 200/cm^2, and more preferably 150/cm^2 on each of the sides.

The height of the protrusions must be not less than 0.5 μm.

The lower limit of the height of the protrusions is preferably 0.8 μm, and more preferably 1.0 μm.

Size and height of the protrusions in embodiments of the present invention are values measured by the measurement method described below.

The increase in air permeation resistance described in embodiments of the present invention is the difference between air permeation resistance of a polyolefin porous membrane that is a substrate and air permeation resistance of a battery separator laminated with a modified porous layer, and preferably not more than 100 sec/100 cc Air.

The overview of the polyolefin porous membrane and the battery separator of embodiments of the present invention is described, but the scope of the invention is not limited to these representative examples.

First, the polyolefin porous membrane of embodiments of the present invention is described.

The thickness of the polyolefin porous membrane of embodiments of the present invention is preferably not greater than 25 μm. The upper limit is preferably 20 μm, and more preferably 16 μm.

The lower limit of thickness is preferably 7 μm, and more preferably 9 μm.

When the thickness of the polyolefin porous membrane is in the preferable ranges described above, practical membrane strength and pore blocking function can be achieved, the area in the unit volume of the battery case is not restricted and it is suitable for increases in battery capacity expected in the future.

The upper limit of the air permeation resistance of the polyolefin porous membrane is preferably 300 sec/100 cc Air, more preferably 200 sec/100 cc Air, and even more preferably 150 sec/100 cc Air, while the lower limit is preferably 50 sec/100 cc Air, more preferably 70 sec/100 cc Air, and even more preferably 100 sec/100 cc Air.

The upper limit of the porosity of the polyolefin porous membrane is preferably 70%, more preferably 60%, and even more preferably 55%, and the lower limit is preferably 30%, more preferably 35%, and even more preferably 40%.

When air permeation resistance and porosity are in the preferable ranges described above, sufficient battery charge/discharge characteristics, especially sufficient ion permeability (charge/discharge operating voltage) and life of a battery (related to the amount of an electrolytic solution to be retained) are sufficient, and functions as a battery are sufficiently exhibited.

Furthermore, shorting is less likely to occur during charge/discharge because sufficient mechanical strength and insulating properties are obtained.

Since the average pore diameter in the polyolefin porous membrane greatly affects pore blocking characteristics, it is preferably from 0.01 to 1.0 μm, more preferably from 0.05 to 0.5 μm, and even more preferably from 0.1 to 0.3 μm.

When the average pore size of the multi-layer polyolefin porous membrane is in the preferable ranges described above, sufficient 0° peel strength of the modified porous layer described above can be achieved due to the anchor effect of the functionalized resin.

Furthermore, significant deterioration of the air permeation resistance upon laminating the modified porous layer can be prevented, the response of the pore blocking phenomenon to the temperature does not slow down, and pore blocking temperature associated with the rate of temperature increase does not shift to a higher temperature.

For the polyolefin resin that constitutes the polyolefin porous membrane, polyethylene and polypropylene are preferable.

Additionally, the polyolefin resin can be a single component, a mixture of two or more kinds of different polyolefin resins, for example a mixture of polyethylene and polypropylene, or a copolymer of different olefins.

They are preferable because they possess pore blocking effects that blocks electrical current and prevents excessive temperature increase upon abnormal temperature increase in batteries, in addition to basic properties such as electrical insulating properties, ion permeability and the like.

Among these, polyethylene is more preferable from the point of view of good pore blocking characteristics.

Details are discussed below with polyethylene as a polyolefin resin used in embodiments of the present invention.

Polyethylene includes ultrahigh molecular weight polyethylene, high density polyethylene, medium-density polyethylene, low density polyethylene and the like.

Additionally, polymerization catalysts are not limited, and include Ziegler-Natta catalyst, Phillips catalyst, metallocene catalyst and the like.

These polyethylenes may be not only a homopolymer of ethylene, but also a copolymer that contains a small amount of another α-olefin.

Preferable, an α-olefin other than ethylene includes propylene, 1-butene, 1-pentene, 1-hexene, 4-methyl-1-pentene, 1-octene, (meth)acrylic acid, ester of (meth)acrylic acid, styrene and the like.

Polyethylene may be a single component, but preferably a mixture of two or more kinds of polyethylene.

As a polyethylene mixture, a mixture of two or more kinds of ultrahigh molecular weight polyethylene with different weight average molecular weight (Mw), or a mixture of similar kind of high density polyethylene, medium-density polyethylene and low density polyethylene can be used. Also, a mixture of two or more kinds of polyethylene selected from the group consisting of ultrahigh molecular

weight polyethylene, high density polyethylene, medium-density polyethylene and low density polyethylene can be used.

As a polyethylene mixture, a mixture of ultrahigh molecular weight polyethylene with Mw not less than 5×10^5 and polyethylene with Mw not less than 1×10^4 and less than 5×10^5 is preferred.

Mw of the ultrahigh molecular weight polyethylene is preferably from 5×10^5 to 1×10^7, more preferably from 1×10^6 to 15×10^6, and most preferably from 1×10^6 to 5×10^6.

As polyethylene with Mw not less than 1×10^4 and less than 5×10^5, any one of high density polyethylene, medium-density polyethylene, or low density polyethylene can be used, but specifically high density polyethylene is preferably used.

As polyethylene with Mw not less than 1×10^4 and less than 5×10^5, two or more kinds of polyethylene with different Mw may be used or two or more kinds of polyethylene with different densities can be used.

Setting the upper limit of Mw of the polyethylene mixture to not greater than 15×10^6 facilitates melting and extruding.

In embodiments of the present invention, the upper limit for ultrahigh molecular weight polyethylene content is preferably 40 wt. %, more preferably 30 wt. %, and even more preferably 10 wt. %. The lower limit is preferably 1 wt. %, more preferably 2 wt. %, and even more preferably 5 wt. %.

If the content of ultrahigh molecular weight polyethylene is within the preferable range, protrusions with sufficient height can be obtained.

These protrusions function as anchors upon laminating a modified porous layer, resulting in exceptionally large peel resistance against a force applied parallel to the planar directions of the polyethylene porous membrane.

Also, even when the polyethylene porous membrane thickness is reduced, sufficient tensile strength can be obtained.

Preferably, tensile strength is not less than 100 MPa.

The upper limit is not particularly defined.

The inventors of the present invention consider the mechanism of protrusion formation as follows.

Crystallization of polyethylene begins simultaneous to extrusion of a resin solution comprising melt polyethylene resin and a forming solvent through the T-die, and the rate of crystallization increases due to rapid quenching caused by contact with the chill-roll.

At this point, spherulites with symmetrical structure containing nuclei are formed (FIG. **2**).

When the heat conduction rate between chill-roll surface and the melt polyethylene resin is relatively small, the rate of crystallization is small, resulting in spherulites containing relatively small nuclei.

If heat conduction rate is large, spherulites containing relatively large nuclei are formed.

These nuclei in the spherulites form protrusions during stretching in TD (transverse direction) and/or MD (machine direction) in the later process.

Meanwhile, spherulites appear as ring-shaped marks on polyethylene porous membrane surface (FIG. **3**).

The ratio (molecular weight distribution Mw/Mn) of weight average molecular weight (Mw) to number-average molecular weight (Mn) of the polyethylene resin is preferably in the range from 5 to 200, and more preferably in the range from 10 to 100.

When Mw/Mn is within the preferable range described above, a sufficient number of protrusions is obtained, and furthermore, sufficient mechanical strength is obtained even when the polyethylene porous membrane thickness is decreased.

The value Mw/Mn is used as a measure of molecular weight distribution, and when this value is larger for polyethylene formed of a single component, the width of molecular weight distribution is larger.

The value Mw/Mn of polyethylene formed of a single component can be adjusted as appropriate by multi-step polymerization of polyethylene.

Meanwhile, the value Mw/Mn of a polyethylene mixture can be adjusted as appropriate by adjusting molecular weight or mixing ratio of each component.

To manufacture a polyethylene porous membrane, as long as the various characteristics described above are satisfied, any manufacturing method suitable for an objective can be selected optionally.

Manufacturing methods of porous membranes include the foaming method, the phase-separation method, the melt-recrystallization method, the drawing-pore-opening method, the powder-sintering method and the like, and the phase-separation method is preferred among them from the viewpoints of homogenization of micropores and costs.

Examples of the manufacturing method according to the phase-separation method include a method comprising heat-melting and kneading, for example, polyethylene with a forming solvent, extruding the resultant molten mixture through a die, cooling the extrudate to form a gel-like product, stretching the obtained gel-like product in one or more axial directions, and removing the forming solvent, to obtain a porous membrane.

The polyethylene porous membrane may be a monolayer membrane or a membrane comprising two or more layers with different molecular weights or different average micropore sizes.

Manufacturing methods of a multi-layer membrane with two or more layers can be any one of the following methods. For example, in one method, each polyethylene that configures layer a and layer b is melt-kneaded with a forming solvent and the obtained molten mixture is extruded from the respective extruders to a die, merging respective gel-sheets configuring each ingredient by co-extrusion. In another method, respective gel-sheets configuring each layer are overlapped and hot-melted.

The co-extrusion method is preferred because a high inter-layer adhesive strength is easily achieved; high permeability is easily maintained because communication holes are easily formed between layers; and productivity is superior.

For a layer configuration with two or more layers, it is preferable that the molecular weight and the molecular weight distribution of the polyethylene resin in one or more outer layer satisfy the conditions described above.

It is necessary for a polyethylene porous membrane to possess the function in which the pores are blocked in the case of abnormal charge/discharge reactions.

Therefore, the melting point (softening point) of the configured resins is preferably from 70 to 150° C., more preferably from 80 to 140° C., and even more preferably 100 to 130° C.

If the melting point of the configured resin is within the preferred range described above, erroneous battery shutdown due to pore blocking function during the normal operation can be avoided and safety is ensured by the pore blocking function in the event of abnormal reaction.

Embodiments of a method of manufacturing the polyolefin porous membrane of the present invention is described.

Embodiments of a method of manufacturing the polyolefin porous membrane of the present invention comprises the steps of:

(a) preparing polyolefin resin solution by melting and kneading after adding a forming solvent to polyolefin resin;

(b) extruding the polyethylene resin solution through a T-die, chilling by a chill-roll having a surface from which the forming solvent has been removed, and disposed on both sides of the polyethylene resin solution that has been extruded in membrane form, and forming a gel-like product;

(c) stretching the gel-like product in MD (the machine direction) and in TD (the transverse direction) to obtain a stretched and molded material;

(d) extracting and removing the forming solvent from the stretched and molded material followed by drying, to obtain a porous molded material;

(e) thermal treating the porous molded material to obtain a polyolefin porous membrane.

Furthermore, the corona treatment step and the like can be optionally implemented after the steps from (a) to (e).

Each step is described below using the example in which polyethylene resin is used as polyolefin resin.

(a) Step of preparing polyethylene resin solution by melting and kneading after adding a forming solvent to polyethylene resin

A forming solvent is not limited as long as it can dissolve polyethylene sufficiently.

For example, aliphatic or cyclic hydrocarbon such as nonane, decane, undecane, dodecane, liquid paraffin and the like, or mineral oil fractions with their boiling points corresponding to these are included. A non-volatile solvent such as liquid paraffin is preferable to obtain a gel-like product with constant solvent content.

Heat-melting is performed by stirring the polyethylene composition or homogeneously mixing polyethylene composition in the extruder at the temperature at which the polyethylene composition can be completely dissolved.

Though the temperature is dependent on polymer and solvent used in case of stirring the polyethylene composition in the solvent or in the extruder, it is preferable to be in the range from 140 to 250° C., for example.

The concentration of the polyethylene resin is preferably from 25 to 40 parts by weight, and more preferably from 28 to 35 parts by weight, per 100 parts by weight of the total of the polyethylene resin and the forming solvent.

If the polyethylene resin concentration is within the preferable range described above, a sufficient number of the nuclei to form protrusions are formed, resulting in a sufficient number of protrusions.

Additionally, swell and neck-in phenomena at the outlet of the T-die can be suppressed as the polyethylene resin solution is extruded, and moldability and self-supporting characteristics of the extrudate are maintained.

The method of melting and kneading is not particularly limited, but normally homogeneous kneading is done in the extruder.

This method is suitable for preparing the solution with high concentration of polyethylene.

The melting temperature is preferably in the range from E+10° C. to E+100° C., where E is the melting point of polyethylene.

Generally, the melting temperature is preferably in the range from 160 to 230° C., and more preferably in the range from 170 to 200° C.

The melting point herein is the value obtained by differential scanning calorimetry (DSC) according to JIS K 7121.

The forming solvent may be added before kneading or added during kneading from the middle of the extruder and then further melt-kneaded. Preferably the forming solvent is added and put in solution in advance before kneading.

For melting and kneading, antioxidant is preferably added to prevent oxidation of polyethylene.

(b) Step of extruding the polyethylene resin solution through a T-die, chilling by a chill-roll having a surface from which the forming solvent has been removed, and disposed on both sides of the polyethylene resin solution that has been extruded in membrane form, and forming a gel-like product

A melt-kneaded polyethylene resin solution is extruded through a T-die from the extruder directly or via another extruder.

As the T-die, a T-die for a sheet with a rectangular mouthpiece is generally used (in some cases denoted as simply "die" in the present specification).

Both sides of the polyethylene resin solution extruded in membrane form from the T-die are brought into contact with a pair of rolling chill-rolls set to a temperature from 20° C. to 40° C. by a refrigerant, thereby forming a gel-like product.

The extruded polyethylene resin solution is preferably cooled down to the temperature not more than 25° C.

The cooling rate in the temperature range at which the crystallization substantially occurs is important.

For example, the extruded polyethylene resin solution is cooled at a cooling rate not less than 10° C./sec in the temperature range at which the surface of the polyethylene resin solution substantially crystallizes, and the gel-like product is obtained.

The cooling rate is preferably not less than 20° C./sec, more preferably not less than 30° C./sec, and even more preferably not less than 50° C./sec.

The cooling step described above can immobilize microphase-separated structures in which a polyethylene phase is separated by solvent, form spherulites comprising relatively large nuclei on the surface of the gel-like product in contact with the chill-roll and form protrusions with appropriate shapes after stretching.

The cooling rate can be estimated by carrying out a simulation using parameters such as extruding temperature of the gel-like product, thermal conductivity of the gel-like product, the thickness of the gel-like product, and heat transfer coefficients of the forming solvent, the chill-roll and air.

Additionally, in embodiments of the present invention, it is important to remove a forming solvent deposited on the chill-roll surface, which will be in contact with the polyethylene resin solution extruded from the T-die, as much as possible.

Namely, as illustrated in FIG. **4**, the polyethylene resin solution is chilled and formed into the gel-like product upon contact with the rolling chill-roll and the forming solvent is present on the chill-roll surface after the formed gel-like product is removed. Generally, the roll surface as-is will be in contact with the polyethylene resin solution again.

However, the large amount of the forming solvent deposited on the chill-roll surface has thermal insulation effect, which decelerates the cooling rate, and hinders the protrusion formation.

Therefore, it is important to remove a forming solvent as much as possible before the chill-roll will be brought into contact with the polyethylene resin solution again.

It is preferable that the diameters of the two chill-rolls disposed on the two sides of the polyethylene resin solution are different.

Furthermore, it is preferable that the placement positions of the rotating shafts of the two chill-rolls relative to the height of the placement position of the polyethylene resin solution outlet of the T-die are different, and moreover, that the rotating shaft of the smaller-diameter chill-roll is closer to the polyethylene resin solution outlet of the T-die than the larger-diameter chill-roll.

This is in order to minimize the long distance between the position of the polyethylene resin solution outlet of the T-die and the polyethylene resin solution contact position on the chill-roll.

For example, the arrangement illustrated in FIG. **4** enables a cooling rate of not less than 10° C./sec in the temperature range at which the polyethylene resin solution extruded from the T-die substantially crystallizes.

The means to remove a forming solvent, namely, the means to remove the forming solvent from the chill-roll is not particularly limited. But the preferable means comprises placing a doctor blade on the chill-roll parallel to the transverse direction of the gel-like product and scraping off the forming solvent to the extent that the forming solvent may not be observable on the chill-roll surface from the point immediately after the doctor blade to the point of gel-like product contact.

Alternatively, the forming solvent can be removed by the means comprising blowing off the forming solvent by compressed air, the means comprising pumping out the forming solvent, or the combination of these means.

Among these means, the means using a doctor blade scraping off the forming solvent is preferable since it is relatively easily implemented, and the use of the two or more doctor blades instead of one is more preferred due to improvement in the effective removal of the forming solvent.

The material of the doctor blade is not particularly limited as long as it has resistance to a forming solvent. But the resin or rubber doctor blade is preferred compared to the metal doctor blade.

The metal doctor blade may possibly damage the chill-roll.

The resin doctor blades include polyester, polyacetal, polyethylene doctor blades and the like.

Even if the chill-roll temperature is set to less than 20° C., the insulating effect of the forming solvent may not only fail to achieve sufficient cooling rate, but also cause roughening of the surface of the gel-like product due to dew formation on the chill-roll.

The thickness of the polyethylene resin solution upon extrusion is preferably not more than 1,500 μm, more preferably not more than 1,000 μm, and even more preferably not more than 800 μm.

If the thickness of the polyethylene resin solution upon extrusion is within the range described above, it is preferable since the cooling rate of the chill-roll surface is not decelerated.

(c) The step of stretching the gel-like product in MD (the machine direction) and in TD (the transverse direction) to obtain a stretched and molded material

Next, the gel-like product is stretched to form a stretched and molded material.

The step of stretching involves heating of the gel-like product and stretching the gel-like product in MD and TD at the pre-determined percentages using the usual tenter method, roll method or combination thereof.

Stretching may be simultaneous stretching (simultaneous biaxial stretching) in MD and TD or may be sequential stretching.

The order of the sequential stretching can be any of the ordering between MD and TD, and stretching in MD and/or TD can be multiple steps.

The stretching temperature is not more than M+10° C., where M is the melting point of the polyolefin composition.

Additionally, the percentage of the stretching is dependent on the thickness of the material, but the planar stretching percentage is preferably not less than 900% and more preferably from 1,600% to 40,000%.

In case of simultaneous stretching in MD and TD (simultaneous biaxial stretching), an identical stretching percentage for MD and TD, such as 300%×300%, 500%×500%, and 700%×700%, is preferred.

The planar stretching percentage within the preferable range described above provides sufficient stretching and results in a porous membrane with high modulus and large strength.

Also, by controlling the stretching temperature, the desirable air permeation resistance can be obtained.

(d) The step of extracting and removing the forming solvent from the stretched and molded material followed by drying, to obtain a porous molded material

In this step, the stretched and molded material is treated with washing solvent to remove residual forming solvent and the porous membrane is obtained.

The washing solvents that can be used are volatile solvents, and include hydrocarbons such as pentane, hexane, heptane and the like, chlorinated hydrocarbons such as methylene chloride, carbon tetrachloride and the like, fluorinated hydrocarbons such as trifluoroethane and the like, ethers such as diethylether, dioxane and the like.

These washing solvents are appropriately selected according to the forming solvent used for dissolving polyethylene, and can be used alone or in combination thereof.

The washing method includes immersion and extraction in the washing solvent, showering the washing solvent, suctioning the washing solvent from the opposite side of the stretched and molded material and the combination thereof.

Washing described above is continued until the residual solvent in stretched and molded material reaches less than 1 wt. %.

After the washing, the washing solvents are dried. The method of drying the washing solvents include heat drying, air-dry and the like.

(e) The step of thermal treating the porous molded material to obtain a polyethylene porous membrane

The porous molded material obtained by drying further undergoes thermal treatment to obtain a polyethylene porous membrane.

The thermal treatment temperature is preferable from 90 to 150° C.

The thermal treatment temperature within the preferable range described above can achieve sufficient reduction in thermal shrinkage and air permeation resistance of the polyolefin porous membrane obtained.

The residence time of the thermal treatment step is not particularly limited, but generally is not less than 1 second and not more than 10 minutes, and preferably from 3 seconds to not more than 2 minutes.

The thermal treatment can be done by any of the methods among the tenter method, roll method, pressing method, or free method.

From the perspective of thermal shrinkage, it is preferable to shrink the porous molded material in at least one of MD

and TD, while the porous molded material is fixed in both directions in the thermal treatment step.

The ratio of shrinking in at least one of MD and TD is preferably from 0.01 to 50%, and more preferably from 3 to 20%.

The ratio of shrinking within the preferable range described above improves the heat shrinkage after 8 hours at 105° C. and maintains air permeation resistance.

Meanwhile, the functionalization steps such as corona treatment step or hydrophilization step can be implemented optionally after the steps (a) to (e).

The modified porous layer used in embodiments of the present invention is described below. The modified porous layer is preferably laminated on both sides of the polyolefin porous membrane.

When modified porous layers are disposed on both sides, it is preferable that the modified porous layers do not easily delaminate even when strong parallel stress is applied to either side due to contact with rollers, bars, or the like during the later process such as slit processing and conveyance process, since the effect of embodiments of the present invention can be demonstrated effectively.

A modified porous layer described in embodiments of the present invention is not particularly limited if the layer includes functionalized resin, namely, resin that provides or improves at least one of the functions among heat resistance, adhesion to an electrode material, wettability toward electrolytic solution and the like. The modified porous layers preferably include inorganic particles or cross-linked polymeric particles in addition to functionalized resin.

As functionalized resin, from the viewpoint of heat resistance improvement, for example, the heat-resistant resin used preferably has a glass transition temperature or melting point not less than 150° C., more preferably not less than 180° C., and even more preferably not less than 210° C.

It is not particularly necessary to specify the upper limits for glass transition temperature or melting point.

If the glass transition temperature is higher than degradation temperature, the degradation temperature may be within the range described above.

If the lower limit of the glass transition temperature is within the preferable range described above, sufficient heat-resistant membrane puncture temperature can be established and the great degree of safety is maintained.

As heat-resistant resin, for example, resin which comprises polyamideimide, polyimide, or polyamide as a major ingredient can be used preferably.

Among these, the resin comprising polyamideimide as a major ingredient is particularly preferred. These resin can be used alone or in combination with the other materials.

From the viewpoint of adhesion to electrodes, it is preferable to use one or more kinds of resin selected from the group consisting of vinylidene fluoride homopolymer, vinylidene fluoride/olefin fluoride copolymer, vinyl fluoride homopolymer and vinyl fluoride/olefin fluoride copolymer, as functionalized resin.

Polytetrafluoroethylene is particularly preferred.

These polymers can sustain sufficient affinity toward electrolytic solution during use at high temperature, since they have sufficient adhesion to electrodes, great affinity toward non-aqueous electrolytic solution, appropriate heat resistance and great chemical and physical stability toward non-aqueous electrolytic solution.

Polyvinylidene fluoride (PVDF) is preferable.

From the perspectives of the environment and cost, water-soluble resin or water-dispersible resin is preferred.

Specific examples include carboxymethylcellulose (CMC), polyvinyl alcohol, and acrylic-based resins such as polyacrylic acid, polyacrylamide, and polymethacrylic acid. CMC and acrylic-based resins are most preferred.

As the acrylic-based resin, commercially available acrylic emulsions may be used, specifically Acryset® TF-300 (manufactured by Nippon Shokubai Co., Ltd.) and Polysol® AP-4735 (manufactured by Showa Denko K.K.).

The details of the functionalized resin are described below using polyamideimide resin as an example.

In general, the synthesis of polyamideimide resin includes normal methods such as the acid chloride method, in which trimellitic chloride and diamine are used, the diisocyanate method, in which trimellitic anhydride and diisocyanate are used. The diisocyanate method is preferable from the viewpoint of manufacturing cost.

An acid component used for the synthesis of polyamideimide resin includes trimellitic anhydride (chloride) and some parts of the acid component may be substituted with the other polybasic acid or anhydride thereof.

Examples include tetracarboxylic acids and anhydrides thereof such as pyromellitic acid, biphenyl tetracarboxylic acid, biphenylsulphone tetracarboxylic acid, benzophenone tetracarboxylic acid, biphenylether tetracarboxylic acid, ethyleneglycol bis-trimellitate, propyleneglycol bis-trimellitate and the like, aliphatic dicarboxylic acids such as oxalic acid, adipic acid, malonic acid, sebacic acid, azelaic acid, dodecane dicarboxylic acid, dicarboxy polybutadiene, dicarboxy poly(acrylonitrile-butadiene), dicarboxy poly(styrene-butadiene) and the like, alicyclic dicarboxylic acids such as 1,4-cyclohexane dicarboxylic acid, 1,3-cyclohexane dicarboxylic acid, 4,4'-dicyclohexyl methane dicarboxylic acid, dimer acid and the like, aromatic dicarboxylic acids such as terephthalic acid, isophthalic acid, diphenylsulphone dicarboxylic acid, diphenyether dicarboxylic acid, naphthalene dicarboxylic acid and the like.

Among these examples, 1,3-cyclohexane dicarboxylic acid and 1,4-cyclohexane dicarboxylic acid are preferable from the viewpoints of electrolytic solution resistance, and dimer acid, dicarboxy polybutadiene with molecular weight not less than 1,000, dicarboxy poly(acrylonitrile-butadiene) and dicarboxy poly(styrene-butadiene) are preferable from the viewpoint of shut-down characteristics.

Also, a urethane group may be introduced into the molecule by substituting the part of trimellitic acid compound with glycol.

The glycol includes alkylene glycol such as ethylene glycol, propylene glycol, tetramethylene glycol, neopentyl glycol, hexandiol and the like, polyalkylene glycol such as polyethyleneglycol, polypropylene glycol, polytetramethylene glycol and the like, polyester with hydroxy terminal group synthesized from one or more than two kinds of dicarboxylic acids described above and one or more than two kinds of glycol described above and the like. Among these, polyethyleneglycol and polyester with hydroxy terminal group are preferred from the viewpoint of shut-down effect.

In addition, number-average molecular weight of these is preferably not less than 500, and more preferably not less than 1,000.

The upper limit is not particularly limited, but preferably it is less than 8000.

If a part of the acid component is substituted with at least one of the group consisting of dimer acid, polyalkylene ether, polyester, and butadiene-based rubber containing any one of carboxyl group, hydroxyl group and amino group at

the terminal, it is preferable to substitute the amount from 1 to 60 mol % of the acid component.

A diamine (diisocyanate) component that is used for synthesis of the polyamideimide resin preferably includes o-tolidine and tolylenediamine as the components, and a component that substitutes the part thereof includes aliphatic diamine such as ethylenediamine, propylenediamine and hexamethylenediamine and diisocyanates thereof, alicyclic diamine such as 1,4-cyclohexanediamine, 1,3-cyclohexanediamine and dicyclohexylmethanediamine and diisocyanates thereof, and aromatic diamine such as m-phenylenediamine, p-phenylenediamine, 4,4'-diaminodiphenyl methane, 4,4'-diaminodiphenyl ether, 4,4'-diaminodiphenyl sulfone, benzidine, xylylenediamine and naphthalene diamine and diisocyanates thereof, and among them, dicyclohexylmethanediamine and diisocyanate thereof are most preferable, and 4,4'-diaminodiphenyl methane, naphthalenediamine and diisocyanates thereof are preferable, from the viewpoints of reactivity, costs and electrolytic solution resistance.

In particular, o-tolidine diisocyanate (TODD, 2,4-tolylene diisocyanate (TDI) and blends thereof are preferable.

Furthermore, for improving 0° peel strength of the modified porous layer, the content of o-tolidine diisocyanate (TODI) that has high stiffness is not less than 50 mol %, preferably not less than 60 mol %, and even more preferably not less than 70 mol %, based on the total amount of all isocyanates.

The polyamideimide resin can be easily produced by stirring in a polar solvent such as N,N'-dimethylformamide, N,N'-dimethylacetamide, N-methyl-2-pyrrolidone, or α-butyrolactone while heating at from 60 to 200° C.

In this case, amine such as triethylamine or diethylene triamine, or an alkali metal salt such as sodium fluoride, potassium fluoride, cesium fluoride or sodium methoxide can also be used optionally as a catalyst.

When the polyamideimide resin is used in embodiments of the present invention, the logarithmic viscosity thereof is preferably not less than 0.5 dL/g.

If the logarithmic viscosity of the polyamideimide resin is within the preferable range described above, the melt-down characteristics can be sufficiently achieved and the polyolefin porous membrane can be prevented from becoming fragile.

Also, 0° peel strength can be improved due to the anchor effect.

Meanwhile, the upper limit is preferably less than 2.0 dl/g, considering the processability and solubility toward solvents.

The modified porous layer of embodiments of the present invention can be obtained by coating a resin solution (referred to as varnish in some cases, hereafter) dissolved in a solvent, which can dissolve functionalized resin and is miscible with water, on a predetermined substrate, phase-separating the functionalized resin and the solvent under the humidified condition, and solidifying the resin under water bath (referred to as a coagulating bath in some cases, hereafter).

A phase-separation assisting agent can be added to varnish optionally.

Solvents to dissolve the functionalized resin include N,N-dimethyl acetamide(DMAc), N-methyl-2-pyrrolidone (NMP), hexamethylphosphoric triamide (HMPA), N,N-dimethylformamide (DMF), dimethyl sulfoxide (DMSO), γ-butyrolactone, chloroform, tetrachloroethane, dichloroethane, 3-chloro naphthalene, parachlorophenol, tetralin, acetone, acetonitrile and the like, and the solvent can be selected according to the solubility of the resin without restriction.

Solid concentration of the varnish is not particularly limited as long as it can be coated uniformly, but it is preferably not less than 50 wt. % and not more than 98 wt. % and more preferably not less than 80 wt. % and not more than 95 wt. %.

The solid concentration of the varnish within the preferable range described above can prevent the modified porous layer from becoming fragile and achieve sufficient 0° peel strength of the modified porous layer.

The Phase-separation assisting agent used in embodiments of the present invention is one or more kinds of agents selected from water, alkylene glycol such as ethylene glycol, propylene glycol, tetramethylene glycol, neopentyl glycol, hexandiol and the like, polyalkylene glycol such as polyethylene glycol, polypropylene glycol, polytetramethylene glycol and the like, water soluble polyester, water soluble polyurethane, polyvinylalcohol, carboxymethylcellulose and the like.

The amount of phase-separation assisting agent added is preferably from 10 to 90 wt. %, more preferably from 20 to 80 wt. %, and even more preferably from 30 to 70 wt. % based on the weight of the varnish solution.

Mixing these phase-separation assisting agents to the varnish can mainly control the air permeation resistance, the surface open porosity and the rate of layer structure formation.

If the added amount of phase-separation assisting agent is within the preferable range described above, the rate of phase separation is significantly increased.

In addition, it will prevent the resin solution from becoming turbid and resin component from deposited during mixing.

In order to reduce the curl of the polyolefin porous membrane caused by laminating the modified porous layer, it is important to add inorganic particles or cross-linked polymeric particles to the varnish.

Addition of inorganic particles or cross-linked polymeric particles to the varnish can prevent internal shorting caused by growth of dendrite crystals of the electrodes inside the battery (anti-dendrite formation effect), reduce the thermal shrinkage and provide lubrication.

The upper limit for the amount of the particles added is preferably 98 wt. % and more preferably 95 wt. %.

The lower limit is preferably 10 wt. % and more preferably 50 wt. %.

When the amount of the particles added is within the preferable range described above, the reduction of the curl is sufficient, the ratio of the functionalized resin based on the total volume of the modified porous layer is the most suitable and sufficient 0° peel strength of the modified porous layer can be obtained.

Examples of inorganic particles include calcium carbonate, calcium phosphate, amorphous silica, crystalline glass fillers, kaolin, talc, titanium dioxide, alumina, silica-alumina conjugate oxide particles, barium sulfate, calcium fluoride, lithium fluoride, zeolite, molybdenum sulphide, mica and the like.

Examples of heat-resistant cross-linked polymeric particles include cross-linked polystyrene particles, cross-linked acrylate-based resin particles, cross-linked methylmethacrylate based particles and the like.

The average size of the particles is preferably not less than 1.5 times and not more than 50 times as large as the average pore diameter of the polyolefin porous membrane.

It is more preferably not less than 2.0 times and not more than 20 times as large as the average pore diameter.

When the average size of the particles is within the preferable range described above, the pores in the polyolefin porous membrane may be blocked while the heat-resistant resin and the particles coexist, and air permeation resistance can be maintained as a result. Meanwhile, it can prevent shedding of particles during the battery assembly process, which may cause serious failure of the battery.

The shape of the particles includes spherical shape, substantially spherical shape, plate shape, needle shape, or polyhedron shape. But it is not particularly limited.

The thickness of the modified porous layer is preferably from 1 to 5 μm, more preferably from 1 to 4 μm, and even more preferably from 1 to 3 μm.

When the thickness of the modified porous layer is within the preferable range described above, a battery separator obtained by laminating the modified porous layer on the polyolefin porous membrane can maintain electrical insulation at or above the melting point, and can prevent abnormal reactions due to having a sufficient pore blocking function.

In addition, the volume upon winding can be reduced, thus being suitable for battery capacity increase.

Furthermore, the reduction in the curl can improve productivity in the battery assembly process.

The porosity of the modified porous layer is preferably from 30 to 90%, and more preferably from 40 to 70%.

If the porosity of the modified porous layer is within the preferable range described above, a battery separator obtained by laminating the modified porous layer has low electrical resistance and exhibits increase in electrical current. Also the membrane strength can be maintained.

The upper limit for the total thickness of the battery separator obtained by laminating the modified porous layer is preferably 25 μm, and more preferably 20 μm.

The lower limit is preferably not less than 6 μm and more preferably not less than 7 μm.

When the thickness of the whole battery separator is within the preferable range described above, the battery separator obtained by laminating the modified porous layer can achieve sufficient mechanical strength and electrical insulation.

In addition, capacity reduction can be avoided since the electrode areas filling the battery can be reduced.

The air permeation resistance of the battery separator is one of the most important characteristics and is preferably in a range from 50 to 600 sec/100 cc Air, more preferably from 100 to 500 sec/100 cc Air, and even more preferably from 100 to 400 sec/100 cc Air.

When the air permeation resistance of the battery separator is within the preferable range described above, sufficient electrical insulation can be achieved and clogging, shorting and membrane puncture can be avoided.

In addition, reduction in the membrane electrical resistance can achieve charge/discharge characteristics and battery life characteristics within the range of practical use.

The method of laminating the modified porous layer of the battery separator of embodiments of the present invention is described below.

The method of laminating the modified porous layer on the polyolefin porous membrane may be a method in which the varnish is directly coated onto the polyolefin porous membrane and converted to the modified porous layer (the direct method), or a method in which a substrate film (e.g. polypropylene film or polyester film) which has been coated with the varnish is placed in a specified humidified environment, and the functionalized resin component and the solvent component are phase-separated to result in the modified porous layer, which is then transferred to the polyolefin porous membrane and laminated (the transfer method).

In the case of the direct method, it does not matter if both sides are coated simultaneously or serially and converted to a modified porous layer, or if one side is coated and converted to a modified porous layer and then the remaining side is coated and converted to a modified porous layer.

In embodiments of the present invention, the preferable method for laminating the modified porous layer comprises the steps (i) and (ii) in the manufacturing steps.

Step (i): a step to form a functionalized resin layer on the polyolefin porous membrane, in which coating the varnish containing the functionalized resin and the inorganic particles or the cross-linked polymeric particles on the polyolefin porous membrane is followed by passing it through a low-humidity zone (absolute humidity is not less than 0.5 g/m^3 and less than 6 g/m^3) and a high-humidity zone (absolute humidity is not less than 7.0 g/m^3 and less than 25.0 g/m^3).

Step (ii): a step to produce a battery separator, in which the composite membrane laminated with the functionalized resin layer obtained in the step (i) is immersed in a coagulating bath so that the functionalized resin layer is converted to a modified porous layer, followed by washing and drying.

The further details are described below.

The varnish comprising the main components of a functionalized resin solution, obtained by dissolving the functionalized resin in a solvent that can dissolve the functionalized resin and is miscible in water, and inorganic particles or cross-linked polymeric particles is coated on the polyolefin porous membrane. Then, the varnish-coated polyolefin porous membrane is placed under the specific humidified environment, and the varnish is phase-separated. Then, the functionalized resin is solidified in the water bath (coagulating bath) and the modified porous layer is obtained.

The low-humidity zone in embodiments of the present invention is a zone controlled to an absolute humidity having an upper limit of preferably less than 6 g/m^3, more preferably 4 g/m^3, and even more preferably 3 g/m^3, and having a lower limit of preferably 0.5 g/m^3 and more preferably 0.8 g/m^3.

When the absolute humidity is within the preferable range described above, the phase-separation is sufficiently achieved to become porous and the increase in air permeation resistance is reduced.

In addition, along with phase-separation, the solidification of the resin configuring the modified porous layer is suppressed, penetration of the resin component configuring the modified porous layer into the polyolefin porous membrane can be achieved sufficiently. As a result, the sufficient 0° peel strength of the modified porous layers can be obtained.

Also, when the time required for passing through the low-humidity zone is less than 3 seconds, the phase separation described above is not sufficiently progressed. When the time required for passing through the low-humidity zone is more than 20 seconds, solidification of the resin configuring the modified porous layer is unpreferably progressed.

Next, the coated membrane is passed through the high-humidity zone for not less than 3 seconds but not more than 10 seconds.

The high-humidity zone in embodiments of the present invention is a zone controlled to an absolute humidity having a lower limit of preferably 6 g/m^3, more preferably 7 g/m^3, and even more preferably 8 g/m^3, and having an

upper limit of preferably 25 g/m^3, more preferably 17 g/m^3, and even more preferably 15 g/m^3.

When the absolute humidity is within the preferable range described above, gelation (immobilization) can be sufficiently achieved, excessive penetration of the resin component configuring the modified porous layer into the polyolefin porous membrane may not occur, and the increase in the air permeation resistance can be reduced.

Meanwhile, excessive solidification of the resin component and excessive reduction in penetration of the functionalized resin component into the polyolefin porous membrane can be prevented and sufficient 0° peel strength can be obtained.

Though the temperature condition is not particularly limited, as long as the absolute humidity is within the range described above for both the low-humidity zone and the high-humidity zone, it is preferable not less than 20° C., and not more than 50° C. from the viewpoint of energy savings.

Examples of the method of coating the varnish include a reverse roll coating method, a gravure coating method, a kiss coating method, a roll brush method, a spray coating method, an air knife coating method, a meyer-bar coating method, a pipe doctor method, a blade coating method and a die coating method, and these methods can be used alone or in combination thereof.

In the coagulating bath, the resin component and the particles are solidified in a three-dimensional mesh-like pattern.

The time duration for immersing in the coagulating bath is preferably not less than 3 seconds.

When the time duration for immersing in the coagulating bath is within the preferable range described above, the sufficient solidification of the resin component can be achieved.

Though the upper limit is not particularly limited, 10 seconds is sufficient.

Furthermore, the porous membrane prior to the washing is immersed in the aqueous solution containing the solvent suitable for the functionalized resin in the amount from 1 to 20 wt. %, preferably from 5 to 15 wt. %, subsequently followed by the washing process using pure water and drying process using hot air of temperature not more than 100° C., and the final battery separator can be obtained.

For washing, general methods such as warming, ultrasonic irradiation or bubbling can be used. Furthermore, to maintain the concentration of each bath constant and improve washing efficiency, a method of removing a solution inside the porous membrane between baths is effective.

Specific examples thereof include a method of forcing out a solution within a porous layer by air or an inert gas and a method of physically squeezing out a solution within a membrane by a guide roll.

It is desirable to store the battery separator of embodiments of the present invention in a dry state. However, if storage in an absolute dry state is difficult, the battery separator is preferably subjected to a vacuum drying treatment at the temperature not more than 100° C. just before use.

The battery separator of embodiments of the present invention can be used in nickel-metal hydride batteries, nickel-cadmium batteries, nickel-zinc batteries, silver-zinc batteries, rechargeable batteries such as lithium rechargeable batteries, lithium polymer rechargeable batteries and the like, plastic film capacitors, ceramic capacitors, electric double layer capacitors and the like as a separator, but is preferably used as a separator for lithium ion rechargeable batteries.

The use of a lithium ion rechargeable battery as a separator is described below as an example.

In lithium ion rechargeable batteries, a positive electrode and a negative electrode are laminated with a separator interposed therebetween, and the separator contains an electrolytic solution (an electrolyte).

The structure of the electrode is not particularly limited, and can be a known structure.

For example, an electrode structure in which disc-shaped positive electrode and negative electrode are placed to face each other (coin type), an electrode structure in which planar positive electrodes and negative electrodes are alternately laminated (lamination type), an electrode structure in which band-shaped positive electrode and negative electrode are overlapped and wound (winding type) and the like can be employed.

The positive electrode normally comprises a current collector and a positive-electrode active material layer formed on the surface of the current collector and containing a positive-electrode active material capable of absorbing and desorbing lithium ions.

Examples of the positive-electrode active material include inorganic compounds such as a transition metal oxide, a composite oxide of lithium and a transition metal (lithium composite oxide), a transition metal sulfide and the like.

Examples of the transition metal include V, Mn, Fe, Co, Ni and the like.

Preferred examples of the lithium composite oxide among the positive-electrode active materials include lithium nickel oxide, lithium cobalt oxide, lithium manganese oxide, a lamellar lithium composite oxide having an α-NaFeO2-type structure as matrix and the like.

The negative electrode includes a current collector and a negative-electrode active material layer formed on the surface of the current collector and containing a negative-electrode active material.

Examples of the negative-electrode active material include carbonaceous materials such as natural graphite, artificial graphite, cokes, carbon black and the like.

An electrolytic solution is obtained by dissolving a lithium salt in an organic solvent.

Examples of the lithium salt include LiClO4, LiPF6, LiAsF6, LiSbF6, LiBF4, LiCF3SO3, LiN(CF3SO2)2, LiC(CF3SO2)3, Li2B10Cl10, LiN(C2F5SO2)2, LiPF4(CF3)2, LiPF3(C2F5)3, lower aliphatic carboxylic acid lithium salts, LiAlCl4, and the like.

These can be used alone or in combination of two or more kinds thereof.

Examples of the organic solvent include an organic solvent having a high boiling point and a high dielectric constant such as ethylene carbonate, propylene carbonate, ethylmethyl carbonate, γ-butyrolactone and the like, and an organic solvent having a low boiling point and a low viscosity such as tetrahydrofuran, 2-methyltetrahydrofuran, dimethoxyethane, dioxolane, dimethyl carbonate, diethyl carbonate and the like.

These can be used alone or in combination of two or more kinds thereof.

Particularly, since an organic solvent having a high dielectric constant has a high viscosity, and an organic solvent having a low viscosity has a low dielectric constant, a mixture of each type of an organic solvent is preferably used.

Upon assembling a battery, the separator of embodiments of the present invention can be impregnated with an electrolytic solution to provide the separator with ion permeability.

Usually, an impregnation treatment is carried out by immersing a microporous membrane in an electrolytic solution at normal temperature.

For example, when a cylindrical battery is assembled, first a positive electrode sheet, a separator (composite porous membrane), and a negative electrode sheet are laminated in this order, and the laminate is wound up from one end to form a winding-type electrode element.

Then, this electrode element is inserted into a battery can, and impregnated with the electrolyte solution described above, followed by caulking a battery cap having a safety valve and serving as a positive electrode terminal via a gasket. Thereby a battery can be obtained.

WORKING EXAMPLES

Embodiments of the present invention are described below using examples. However, the present invention is not limited by these examples in any way.

Meanwhile, the measurement values in the working examples are the values measured by the methods described below.

1. Number of Protrusions

The numbers and sizes of the protrusions were measured using a confocal microscope (HD100, manufactured by Lasertec Corporation), placed on a vibration isolation platform, after the light source thereof was stabilized.

Steps

(1) A square frame measuring 1 cm×1 cm was drawn with a fine oily marker pen on any one face (taken as side A) of the battery separator obtained in the working examples and the comparative examples.

(2) The sample was placed on the sample stage with the surface with the drawn square up, and was fixed firmly to the sample stage using the electrostatic adhesion apparatus, the accessory of the confocal microscope.

(3) Using the objective lens with magnification of 5 times, a ring-shaped mark originating from a polyethylene spherulite as in FIG. **3** was displayed on the monitor as a two-dimensional image (called REAL screen for this microscope), and the position of the sample stage was adjusted so that the darkest part of the ring-shaped mark was approximately at the center of the monitor screen.

When two ring-shaped marks were connected, the stage was adjusted so that a point of contact was at the center of the monitor screen.

The protrusion height was measured only for the ring-shaped mark originating from the polyethylene spherulite with its major axes not less than 0.2 mm.

The major axis length of a ring-shaped mark was measured by reading the length between two ends of the ring-shaped mark using a cursor on the two-dimensional image described above.

(4) Changing the objective lens to the one with magnification of 20 times, the reference height (called REFSET in this microscope) was set to the height when the focus was adjusted to the center area of the monitor screen (so that the center area of the monitor screen was displayed most brightly in this microscope).

(5) The measurement range in the height direction was set to 15 μm up and down relative to the reference height as 0 μm described above.

Also, the scan time was 120 seconds, STEP transport distance was 0.1 μm/Step and the three-dimensional data was acquired.

(6) After the acquisition of the three-dimensional data, the image for data processing (Z image in this microscope) was displayed and smoothing process was applied (the conditions for smoothing are: the size of filter: 3×3, matrix type: SMOOTH3_0, the number of process: 1).

In addition, plane correction was applied optionally in the plane correction window as necessary.

(7) The cursor was positioned horizontally on the position intersecting the highest protrusion in the image for data processing (the brightest position) and the cross-sectional profile corresponding to the cursor was displayed on the cross-sectional profile image.

(8) The protrusion size was measured by positioning the two vertical cursors to the inflection points at each side of the skirt of the protrusion in the cross-sectional profile image, and was defined as the distance between the two cursors.

(9) The protrusion height was measured by positioning the two horizontal cursors to the peak of the protrusion and the inflection points at each side of the skirt of the protrusion (the lower inflection point if the two inflection points at each side of the skirt of the protrusion were different in height), and was defined as the distance between two cursors.

(10) The operation described above was repeated inside the square of 1 cm×1 cm described above, and the protrusions with size not less than 5 μm and not more than 50 μm and with height not less than 0.5 μm and not more than 3.0 μm were counted per 1 cm^2 on side A, and the number of the protrusions per 1 cm^2 was determined. Additionally, the average height of those protrusions was determined and taken as the average height of the protrusions on side A.

The same operation was performed for the side opposite side A (taken as side B), and the number of protrusions and the average height of the protrusions on side B were determined.

2. 0° Peel Strength of Modified Porous Layer

To measure any side (for example, side A), the modified porous layer of the opposite side (side B) was delaminated in advance using adhesive tape to expose one surface of the polyolefin porous membrane. The resulting piece was submitted as a sample.

FIG. **1** schematically illustrates the evaluation method.

The reference numeral 1 denotes a laminated sample, 2 denotes a polyolefin porous membrane, 3 denotes a modified porous layer, 4 denotes a double-sided adhesive tape, 5 and 5' denotes aluminum plates, and the arrow in the figure indicates the direction of tensile force.

On the aluminum plate **5**, which was 50 mm long×25 mm wide and 0.5 mm thick, a piece of double-sided adhesive tape **4** (NW-K50, manufactured by Nichiban) of the same size was adhered.

The surface of the polyolefin porous membrane **2** of the sample **1** (the battery separator), cut out in a size of 100 mm long×50 mm wide, was adhered on the aluminum plate **5**, so that the 40 mm portion of the sample, measured from one of the 25-mm-wide edges of the aluminum plate **5**, was overlapped and adhered together, while the protruded portion was cut out.

Next, a double-sided adhesive tape was adhered on one side of the aluminum plate **5'**, 100 mm long×15 mm wide and 0.5 mm thick. The 20 mm portion of the sample, measured from one of the 25-mm-long edges of the aluminum plate **5**, was overlapped and adhered together.

Then, the aluminum plate **5** and the aluminum plate **5'**, which sandwiched the sample, were attached to the tensile tester (Autograph AGS-J1 kN, manufactured by Shimadzu Corp.), and the aluminum plate **5** was affixed. A load was applied on the aluminum plate **5** and the aluminum plate **5'** in parallel but in opposite directions, at a tensile velocity of

10 mm/min, and the strength when the modified porous layer was delaminated was measured.

The measurement was performed for any three points separated not less than 30 cm from each other in the longitudinal direction, and the average value thereof was recorded as the 0° peel strength of the modified porous layer of side A.

The 0° peel strength of the modified porous layer of side B was similarly determined.

3. Membrane Thickness

The membrane thickness was obtained by averaging measurements of 20 points using a contact-type membrane thickness meter (Litematic series 318, manufactured by Mitutoyo Corp.).

An ultrahard spherical contact point of (1)9.5 mm was used and measurements were done at 0.01 N measuring force.

4. Average Pore Diameter

The average pore diameter of the polyolefin porous membrane was measured by the following method.

A sample was fixed on a measuring cell using a double-sided adhesive tape, on which platinum or gold was vacuum-deposited for several minutes, and the SEM measurement was conducted for the surface of the membrane under appropriate magnification.

Any 10 points in an image obtained in the SEM measurement were selected, an average value of pore diameters at the 10 points was determined as an average pore size of the sample.

5. Air Permeation Resistance

Using Gurley Type Densometer Model B manufactured by TESTER SANGYO CO., LTD, the polyolefin porous membrane or the battery separator, which was fixed between a clamping plate and an adapter plate so that no winkle occurred, was measured in accordance with JIS P8117.

A 10 cm-square sample was used, the measurement was performed for a total of 5 points as measurement points, including the center and four corners of the sample. An average value of 5 points was used as air permeation resistance [sec/100 cc Air].

In the case where the length of one side of the sample was less than 10 cm, a value obtained by the measurement for 5 points at intervals of 5 cm may be used.

The increase in the air permeation resistance was obtained by the following formula.

$$\text{The increase in the air permeation resistance} = (Y) - (X) \text{ sec/100 cc Air}$$

The air permeation resistance of the polyolefin porous membrane (X) sec/100 cc Air

The air permeation resistance of the battery separator (Y) sec/100 cc Air

6. Logarithmic Viscosity

A solution obtained by dissolving 0.5 g of heat-resistant resin in 100 mL of NMP was measured at 25° C. using an Ubbelohde viscometer.

7. Melting Point

Using a differential scanning calorimeter (DSC) (DSC6220 manufactured by SII NanoTechnology Corp.), a resin sample of 5 mg was subjected to a temperature increase at the rate of temperature increase of 20° C./min under nitrogen gas atmosphere, and the apex of the melting peak observed during the temperature increase was determined as the melting point.

8. Glass Transition Temperature

A resin solution or a resin solution obtained by dissolving only a modified porous layer by immersing a battery separator in a good solvent was applied onto a PET film (E5001 manufactured by Toyobo Co., Ltd.) or a polypropylene film (Pyrene®-OT manufactured by Toyobo Co., Ltd.) using an applicator with an appropriate gap. A film obtained was peeled off after pre-drying at 120° C. for 10 minutes, then fixed in a metal frame having an appropriate size with a heat-resistant adhesive tape, and further dried in vacuum at 200° C. for 12 hours. The dried film was obtained.

A sample having a width of 4 mm and a length of 21 mm was cut out from the dried film obtained, and the measurement was conducted using a dynamic viscoelasticity measuring apparatus (DVA-220, manufactured by IT Keisoku Seigyo Co. Ltd.) for the measurement length of 15 mm. The measurements were performed for the temperature from room temperature to 450° C., under the conditions of 110 Hz and a rate of temperature increase of 4° C./min. At the inflection point of storage modulus (E'), the glass transition temperature was defined as the intersection of the extrapolated line from the baseline below the glass transition temperature and the tangent at the maximum slope of the curve at the temperature equal to or greater than the inflection point.

9. Porosity

A 10 cm-square sample was prepared, and the sample volume (cm^3) and the mass (g) thereof were measured. The porosity (%) was calculated from the obtained results using the following formula.

$$\text{Porosity} = (1 - \text{mass}/(\text{resin density} \times \text{sample volume})) \times 100$$

10. Scratch Resistance

Both edges of the rolled battery separator obtained from working examples and comparative examples were subjected to slit processing.

Slit processing was carried out by a slitter (model WA177A, manufactured by Nishimura Seisakusho Corp.), under the condition of velocity of 20 m/min and tensile force of 50 N/100 mm.

For the rolls which had contact with the coated surface of one side during the process, two hard chrome plated rolls (both were free rolls) were used.

Then, while winding back the rolled battery separator after slit processing, delaminated defects equal to or larger than 0.5 mm in major axes were counted under visual observation and observation using a magnifying glass of 10 times magnification with a scale (SCALE LUPE×10, manufactured by PEAK). The results were evaluated according to the criteria below.

The evaluation area was taken as 100 mm wide×500 m long (if the width was less than 100 mm, the evaluation length was adjusted so that the evaluation area was the same).

The samples were roll-shaped battery separators removed from the battery separators obtained in the working examples and the comparative examples, and one of the roll-shaped battery separators was measured on side A, while the other roll-shaped battery separator was measured on the opposite side B.

Evaluation Criteria

○ (Exceptionally Good): Not more than 10 on both sides

Δ (Good): 11 to 30 on both sides

X (Poor): 31 or more on at least one side

Working Example 1

A polyethylene composition (melting point 135° C.) was obtained by addition of 0.375 parts by weight of an anti-

oxidant to 100 parts by weight of a composition (Mw/Mn=16.0), which comprises 2 wt. % of ultrahigh molecular weight polyethylene (UHMWPE) with the weight average molecular weight of 2,000,000 and 98 wt. % of high density polyethylene (HDPE) with the weight average molecular weight of 350,000.

30 parts by weight of the polyethylene composition was introduced into a biaxial extruder.

Through the side-feeder of the biaxial extruder, 70 parts by weight of liquid paraffin was supplied, melt and kneaded and the polyethylene resin solution was prepared inside the extruder.

This was followed by extruding the polyethylene resin solution through the T-die disposed at the end of the extruder at 190° C. and at an extruding thickness of 825 μm, disposing the polyethylene resin solution on both sides (refer to FIG. **4**), and drawing by two chill-rolls with internal cooling water temperature kept at 25° C., to form the gel-like product.

During this operation, in each of the chill-rolls, one polyester doctor blade was placed in contact with the chill-roll at the point between the point the gel-like product departing from the chill-roll and the point polyethylene resin solution extruded from the T-die contacting with the chill-roll, parallel to the transverse direction of the gel-like product, and scraped off the liquid paraffin deposited on the chill-roll surface.

Subsequently, the gel-like product was stretched by simultaneous biaxial stretching at 500%×500%, under temperature control to obtain the desired air permeation resistance, and the stretched and molded material was obtained.

The stretched and molded material obtained was washed with methylene chloride to extract and remove residual liquid paraffin, dried and a porous molded material was obtained.

Then, the porous membrane was fixed on a tenter, shrunk by 10% only in TD (the transverse direction), and subjected to thermal treatment at 90° C. for 3 seconds. The polyethylene porous membrane with the thickness of 16 μm, porosity of 45%, average pore diameter of 0.15 μm and air permeation resistance of 240 sec/100 cc Air was obtained.

Synthesis of Heat-Resistant Resin

In a four-necked flask equipped with a thermometer, a condenser and a nitrogen gas introducing tube, 1 mol of a trimellitic anhydride (TMA), 0.8 mol of o-tolidine diisocyanate (TODD, 0.2 mol of 2,4-tolylene diisocyanate (TDI) and 0.01 mol of potassium fluoride were introduced together with N-methyl-2-pyrrolidone to make up the solid concentration of 14%, and the mixture was stirred at 100° C. for 5 hours. Then the mixture was diluted with N-methyl-2-pyrrolidone to the solid concentration of 14% and a polyamideimide resin solution was synthesized.

The logarithmic viscosity of the polyamideimide resin obtained was 1.35 dL/g and the glass transition temperature thereof was 320° C.

The polyamideimide resin solution, alumina particles of the average size of 0.5 μm, and N-methyl-2-pyrrolidone were mixed in the weight ratio of 26:34:40, respectively, introduced into a polypropylene container with zirconia oxide beads (Torayceram® beads, manufactured by Toray Industries, Inc., size 0.5 mm), and dispersed for 6 hours using a paint-shaker (manufactured by Toyo Seiki Seisakusho).

Then, the varnish (a) was obtained by filtering through a filter with filtering limit of 5 μm.

The varnish was coated by gravure coating on both sides of the polyethylene porous membrane, passed through the low-humidity zone at temperature of 25° C. and the absolute humidity of 1.8 g/m^3 for 8 seconds, followed by passing through the high-humidity zone at temperature of 25° C. and the absolute humidity of 12 g/m^3 for 5 seconds. The treated membrane was immersed in an aqueous solution containing 5 wt. % of N-methyl-2-pyrrolidone for 10 seconds.

Then the treated membrane was washed with pure water, dried by passing through the hot-air drier of temperature at 70° C., and the battery separator with final thickness of 18 μm was obtained.

Working Example 2

A battery separator was obtained in the same manner as Working Example 1, except that the compounding ratio of the ultrahigh molecular weight polyethylene (UHMWPE) with the weight average molecular weight of 2,000,000 and the high density polyethylene (HDPE) with the weight average molecular weight of 350,000 was changed to 10:90 (wt. % ratio).

Working Example 3

A battery separator was obtained in the same manner as Working Example 1, except that the compounding ratio of the ultrahigh molecular weight polyethylene (UHMWPE) with the weight average molecular weight of 2,000,000 and the high density polyethylene (HDPE) with the weight average molecular weight of 350,000 was changed to 20:80 (wt. % ratio).

Working Example 4

A battery separator was obtained in the same manner as Working Example 1, except that the compounding ratio of the ultrahigh molecular weight polyethylene (UHMWPE) with the weight average molecular weight of 2,000,000 and the high density polyethylene (HDPE) with the weight average molecular weight of 350,000 was changed to 30:70 (wt. % ratio).

Working Example 5

A battery separator was obtained in the same manner as Working Example 1, except that the compounding ratio of the ultrahigh molecular weight polyethylene (UHMWPE) with the weight average molecular weight of 2,000,000 and the high density polyethylene (HDPE) with the weight average molecular weight of 350,000 was changed to 40:60 (wt. % ratio).

Working Example 6

A battery separator was obtained in the same manner as Working Example 1, except that for both of the two chill-rolls, two polyester doctor blades were disposed to contact with the chill-rolls with the interval of 20 mm.

Working Example 7

A battery separator was obtained in the same manner as Working Example 1, except for both of the two chill-rolls, three polyester doctor blades were disposed to contact with the chill-rolls with the interval of 20 mm from each other.

Working Example 8

The varnish (b) was obtained from polyamideimide resin solution (a), alumina particles with average particle size of

0.5 μm and N-methyl-2-pyrrolidone, with the compounding ratio of 17:43:40 (wt. ratio), respectively.

A battery separator was obtained in the same manner as Working Example 1, except that the varnish (b) was used.

Working Example 9

The varnish (c) was obtained from polyamideimide resin solution (a), alumina particles with average particle size of 0.5 μm and N-methyl-2-pyrrolidone, with the compounding ratio of 39:22:39 (wt. ratio), respectively.

A battery separator was obtained in the same manner as Working Example 1, except that the varnish (c) was used.

Working Example 10

A battery separator was obtained in the same manner as Working Example 1, except the internal cooling water temperature of both of the two chill-rolls was kept at 35° C.

Working Example 11

A battery separator with the final thickness of 22 μm was obtained in the same manner as Working Example 1, except that the polyethylene porous membrane with the thickness of 20 μm was obtained by adjusting the extruding volume of the polyethylene solution.

Working Example 12

A battery separator with the final thickness of 14 μm was obtained in the same manner as Working Example 1, except that the polyethylene porous membrane with the thickness of 12 μm was obtained by adjusting the extruding volume of the polyethylene solution.

Working Example 13

A battery separator with the final thickness of 11 μm was obtained in the same manner as Working Example 1, except that the polyethylene porous membrane with the thickness of 9 μm was obtained by adjusting the extruding volume of the polyethylene solution.

Working Example 14

A battery separator was obtained in the same manner as Working Example 1, except that the 26 parts by weight of polyethylene composition was introduced into the biaxial extruder and 74 parts by weight of liquid paraffin was supplied through the side-feeder of the biaxial extruder.

Working Example 15

A battery separator was obtained in the same manner as Working Example 1, except that the 35 parts by weight of polyethylene composition was introduced into the biaxial extruder and 65 parts by weight of liquid paraffin was supplied through the side-feeder of the biaxial extruder.

Working Example 16

A varnish (d) was obtained by replacing the alumina particles with cross-linked polymeric particles (polymethylmethacrylate-based cross-linked polymeric particles (EPOSTER® MA1002, average particle size 2.5 μm, manufactured by Nippon Shokubai Co. Ltd.,) and the compounding ratio of 35:10:55 (wt. ratio) for cross-linked polymeric particles, N-methyl-2-pyrrolidone.

A battery separator was obtained in the same manner as Working Example 1, except that the varnish (d) was used.

Working Example 17

A fluorine-based resin solution (KF Polymer® #1120, polyvinylidene fluoride (melting point 175° C., 12% N-methylpyrrolidone solution) manufactured by Kureha Chemical Industry Co., Ltd.), alumina particles of average particle size of 0.5 μm, and N-methyl-2-pyrrolidone were mixed at the weight ratio of 31:34:45, respectively. The mixture was introduced into a polypropylene container with zirconium oxide beads (Torayceram® beads, manufactured by Toray Industries, (size 0.5 mm)) and dispersed for 6 hours using a paint-shaker (manufactured by Toyo Seiki Seisakusho).

Then, the varnish (e) was obtained by filtering through a filter with filtering limit of 5 μm.

A battery separator was obtained in the same manner as Working Example 1, except that the varnish (e) was used.

Working Example 18

60.8 parts by mass of a solvent was added to 0.8 parts by mass of CMC (product number 2200, manufactured by Daicel FineChem Ltd.), and this was stirred for 2 hours.

Then, 38.4 parts by mass of substantially spherical alumina fine particles with an average particle size of 0.5 μm were added, and a varnish (f) was obtained by the same method as Working Example 1.

The varnish (f) was coated on both sides of the polyethylene porous membrane by gravure coating and dried by passing through the hot-air drier of temperature 70° C., and a battery separator with a final thickness of 18 μm was obtained.

Working Example 19

48.0 parts by mass of a solvent was added to 6.3 parts by mass of Acryset® TF-300 (manufactured by Nippon Shokubai Co., Ltd.) (solid content 40%), and this was stirred for 2 hours.

Then, 64.2 parts by mass of substantially spherical alumina fine particles with an average particle size of 0.5 μm were added, and a varnish (g) was obtained by the same method as Working Example 1.

A battery separator was obtained in the same manner as Working Example 18, except that the varnish (g) was used.

Comparative Example 1

A battery separator was obtained in the same manner as Working Example 1, except that the liquid paraffin deposited on both of the two chill-rolls was not scraped off by the doctor blade upon obtaining the gel-like product by cooling the polyethylene resin solution extruded from the T-die on the two chill-rolls.

Comparative Example 2

A battery separator was obtained in the same manner as Working Example 1, except that the polyethylene composition (melting point 135° C.) obtained by adding 0.375 parts by weight of an antioxidant to 100 parts by weight of the composition (Mw/Mn=16.0) comprising 100 wt. % of high

density polyethylene (HDPE) with weight average molecular weight of 350,000 was used.

Comparative Example 3

A battery separator was obtained in the same manner as Working Example 1, except the internal cooling water temperature of both of the two chill-rolls was kept at 0° C. and the doctor blade was not used.

Comparative Example 4

A battery separator was obtained in the same manner as Working Example 1, except the polyethylene resin solution extruded from the T-die was immersed in water kept at 25° C. for 1 minute instead of being cooled by the chill-rolls.

Comparative Example 5

50 parts by weight of the polyethylene composition used in the Working Example 1 was introduced into the biaxial extruder, 50 parts by weight of liquid paraffin was supplied through the side-feeder of the biaxial extruder, melt and kneaded and the polyethylene solution was prepared in the extruder. Extrusion of the polyethylene solution from the T-die was tried but a homogeneous membrane was not obtained.

Comparative Example 6

A battery separator was obtained in the same manner as Working Example 1, except the internal cooling water temperature of both of the two chill-rolls was kept at 50° C.

The conditions of Working Examples 1 to 19 and Comparative Examples 1 to 6 are shown in Table 1.

The characteristics of the polyolefin porous membranes and the battery separators obtained from the Working Examples 1 to 19 and Comparative Examples 1 to 6 are listed in Table 2.

TABLE 1

	UHMWPE (wt. %)	HDPE (wt. %)	Resin Concentration (Parts By Weight)	Chill-Roll Temperature (° C.)	Scraping off the Forming Solvent (Number of Blades per Roll)	Varnish
Working Example 1	2	98	30	25	1	a
Working Example 2	10	90	30	25	1	a
Working Example 3	20	80	30	25	1	a
Working Example 4	30	70	30	25	1	a
Working Example 5	40	60	30	25	1	a
Working Example 6	2	98	30	25	2	a
Working Example 7	2	98	30	25	3	a
Working Example 8	2	98	30	25	1	b
Working Example 9	2	98	30	25	1	c
Working Example 10	2	98	30	35	1	a
Working Example 11	2	98	30	25	1	a
Working Example 12	2	98	30	25	1	a
Working Example 13	2	98	30	25	1	a
Working Example 14	2	98	26	25	1	a
Working Example 15	2	98	35	25	1	a
Working Example 16	2	98	30	25	1	d
Working Example 17	2	98	30	25	1	e
Working Example 18	2	98	30	25	1	f
Working Example 19	2	98	30	25	1	g
Comparative Example 1	2	98	30	25	0	a
Comparative Example 2	0	100	30	25	1	a
Comparative Example 3	2	98	30	0	0	a
Comparative Example 4	2	98	30	25° C. (water bath)	—	a
Comparative Example 5	2	98	50	—	—	—
Comparative Example 6	2	98	30	50	1	a

TABLE 2

	Characteristics					
	Polyolefin Porous Membrane Thickness (μm)	Polyolefin Porous Membrane Air Permeation Resistance (X) (sec/100 ccAir)	Air Permeation Resistance (Y) of Battery Separator (sec/100 ccAir)	Air Permeation Resistance Increase [(Y) − (X)] (sec/100 ccAir)	Number of Protrusions of Side A (Per cm^2)	Average Protrusion Height of Side A (μm)
Working Example 1	16	240	308	68	14	2.8
Working Example 2	16	252	309	57	17	1.1
Working Example 3	16	260	317	57	19	0.8
Working Example 4	16	273	325	52	43	0.6

TABLE 2-continued

	Characteristics					
Working Example 5	16	95	146	51	124	0.5
Working Example 6	16	240	308	68	16	2.9
Working Example 7	16	240	309	69	17	2.9
Working Example 8	16	240	312	72	14	2.8
Working Example 9	16	240	307	67	14	2.8
Working Example 10	16	240	339	99	12	2.6
Working Example 11	20	250	319	69	12	2.9
Working Example 12	12	170	238	68	18	1.0
Working Example 13	9	220	280	60	21	0.6
Working Example 14	16	228	290	62	11	2.5
Working Example 15	16	250	314	64	19	2.9
Working Example 16	16	245	314	69	14	2.8
Working Example 17	16	243	317	74	14	2.8
Working Example 18	16	240	310	70	14	2.8
Working Example 19	16	240	310	70	14	2.8
Comparative Example 1	16	239	307	68	0	—
Comparative Example 2	16	238	301	63	0	—
Comparative Example 3	16	240	307	67	0	—
Comparative Example 4	16	241	307	66	2	0.6
Comparative Example 5	—	—	—	—	—	—
Comparative Example 6	16	241	312	71	0	—

	Number of Protrusions of Side B (Per cm^2)	Average Protrusion Height of Side B (μm)	Side A 0° Peel Strength (N/15 mm)	Side B 0° Peel Strength (N/15 mm)	Scratch Resistance
Working Example 1	13	2.6	119	118	○
Working Example 2	16	1.0	117	116	○
Working Example 3	18	0.7	114	113	○
Working Example 4	40	0.6	117	116	○
Working Example 5	120	0.5	103	102	○
Working Example 6	15	2.8	125	124	○
Working Example 7	16	2.7	127	126	○
Working Example 8	13	2.7	114	113	○
Working Example 9	13	2.7	123	122	○
Working Example 10	11	2.5	117	116	○
Working Example 11	11	2.7	117	116	○
Working Example 12	17.0	0.9	115	114	○
Working Example 13	19	0.5	109	107	○
Working Example 14	10	2.4	117	116	○
Working Example 15	18	2.8	121	120	○

TABLE 2-continued

	Characteristics				
Working Example 16	13	2.7	109	107	○
Working Example 17	13	2.7	124	120	○
Working Example 18	13	2.7	125	123	○
Working Example 19	13	2.7	117	116	○
Comparative Example 1	0	—	94	92	X
Comparative Example 2	0	—	92	90	X
Comparative Example 3	0	—	90	88	X
Comparative Example 4	1	0.5	97	95	Δ
Comparative Example 5	—	—	—	—	—
Comparative Example 6	0	—	91	90	X

REFERENCE SIGNS LIST

1 Battery separator
2 Polyolefin porous membrane
3 Modified porous layer
4 Double-sided adhesive tape
5 Aluminum plate
5' Aluminum plate
6 Polyethylene spherulite nucleus
7 T-die
8 Polyolefin resin solution
9 Chill-roll
9' Chill-roll
10 Doctor blade
11 Gel-like product

The invention claimed is:

1. A polyolefin porous membrane comprising protrusions of polyolefin formed at locations corresponding to locations of crystallized spherulites, wherein the protrusions have a size (W) within a range of 5 μm≤W≤50 μm and a height (H) within a range of 0.5 μm≤H, wherein the protrusions are irregularly disposed on opposing sides of the polyolefin porous membrane with a density greater than or equal to $3/cm^2$ per side and less than or equal to $200/cm^2$ per side, and wherein a thickness of the polyolefin porous membrane is less than or equal to 25 μm.

2. A battery separator, comprising:

a polyolefin porous membrane comprising protrusions of polyolefin formed at locations corresponding to locations of crystallized spherulites, wherein the protrusions have a size (W) within a range of 5 μm≤W≤50 μm and a height (H) within a range of 0.5 μm≤H, wherein the protrusions are irregularly disposed on opposing sides of the polyolefin porous membrane with a density greater than or equal to $3/cm^2$ per side and less than or equal to $200/cm^2$ per side, and wherein a thickness of the polyolefin porous membrane is than or equal to 25 μm; and

a modified porous layer laminated on at least one side of the polyolefin porous membrane.

3. The battery separator according to claim **2**, wherein the modified porous layer comprises at least one of a polyamideimide resin, a polyimide resin, and a polyamide resin.

4. The battery separator according to claim **2**, wherein the modified porous layer comprises a fluorine-based resin.

5. The battery separator according to claim **2**, wherein the modified porous layer comprises at least one of carboxymethylcellulose (CMC) and am acrylic-based resin.

6. The battery separator according to claim **2**, wherein the modified porous layer comprises at least one of inorganic particles and cross-linked polymer particles.

7. The battery separator according to claim **2**, wherein the thickness of the polyolefin porous membrane is less than or equal to 20 μm.

8. The battery separator according to claim **2**, wherein the thickness of the polyolefin porous membrane is less than or equal to 16 μm.

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